

Ferdinand W. Santos

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OBJECTIVE

Engineering undergraduate with a minor in Applied Computing seeking an internship to improve my skill set and explore opportunities. I am interested in long-term career development where my interdisciplinary skills can be of use.

EDUCATION

B.S. Interdisciplinary Engineering, San Jose State University Graduated: May 2025

Minor: Applied Computing for Behavioral and Social Science

A.S. Quantitative Reasoning and Scientific Inquiry, San Jose City College Graduated: July 2019

Relevant Coursework: Fall 2024: Computer and System Simulation, Human Interaction

Completed: Physics, Renewable energy engineering, Programming Concepts for Behavioral and Social Science Applications

Math Level: Calculus 3, Differential equations, Probability and Statistics

SKILLS

Technical: Raspberry Pi Pico, Micro python, Thonny, C, C++, Java, Python, R, Pandas, Discord, G-Suite, Microsoft Office

Soft Skills: Analytical, Collaborative, Adaptive, Dedicated, Initiative-taker, Punctual, Reliable

RELEVANT ACADEMIC EXPERIENCE

STEM Core Member August 2017- Present

- Participant in STEM-based programs to advance into major preparation coursework, participate in various conferences, trips, and group activities to further knowledge of STEM fields.
- Acquired knowledge in various engineering principles and actively engaged in hands-on coding projects and collaborative group assignments
- Demonstrated a commitment to continuous learning and skill development

DSPS STEM Tutor San Jose City College August 2017 - 2019

- Provided educational support for the students with disabilities at San Jose City College.
- Provided excellent customer service to the students and collaborated with faculty, staff, and students, while delivering educational support on campus.
- Followed standard operational procedures in a thorough and consistent manner.
- Assisted the student population with requests for reasonable accommodations for academic reasons.

Cyber Defenders, Internship June 2017 - August 2017

- Hybrid Internship with Intro to Java course, includes college credits.
- Assisted with Java and Python programming for software applications.
- Conducted research and development with virtual machines, while creating a virus detection software program.
- Engaged in malware detection research, machine learning, phishing detection best practices, and other network security measures.

PROJECTS

Automated Pill Dispenser: Assisted disabled people to distinguish and sort their medications using rotating sensors into 2 levels of sorting one for correcting and detecting. Utilized Micro python and Thonny paired with a LCD screen to show information in live time and give directions

Covid Computing Project

Created a Chatbot that interacted with the user during the pandemic to help calm nerves and give recommendations.

Information and Statistics regarding Covid was obtained using CSV files, Web Scraping, and User Input gathered by a Python created Chat Bot. Histograms, Bar Charts, Line graphs, Pie charts, and other statistical examples were used to display results.

Online Banking System: ATM services- withdraws and deposits and transfers

Collaborated with multiple engineering disciplines to utilize Html, CSS and JavaScript to build functional software. Database management was required for this project using MySQL to store user accounts and manage transaction data.

Mental health Analysis of the Tech Sector

Gathered CSV files from questionnaires on job satisfaction to create data to use with the R programming language to make histograms and pie charts to story tell with analytics.

Mesh Network Fire Detection System

Designed to improve early wildfire detection by combining environmental sensors with machine learning. Each node collects data on gas levels and light spectra to identify patterns consistent with combustion events. These nodes communicate via a LoRa or Wi-Fi mesh network, sending information to a centralized Gateway Server. The server processes and distributes data through a web dashboard and real-time notification system, allowing users to view flame alerts, sensor activity, and historical trends.

Solar Panel Renewable Energy

Built a Solar Panel using a picture frame and Mini Monocrystalline Solar Cells. The electrical wiring had to be soldered to the solar cells in series to generate power. Data was gathered from the captured solar panel to distinguish solar energy capabilities. Results were documented and a graph was created to show conclusions of the harnessed energy throughout different times and conditions.

WORK EXPERIENCE

CTS Advantage Logistics, Class C Bobtail, Driver

January 2008 - August 2010

- Provided excellent customer service to the clients.
- Safeguarding confidential information and assets for the clients and corporation.
- Collaborated with colleagues, stakeholders, and clients, while meeting corporate objectives.
- Operated a vehicle and engaged in deliveries throughout California.
- Completed a driver's log, documented the contracts, and preserved the records for safekeeping.

K Designers, Construction Associate

December 2005 - January 2008

- Provided excellent customer service to the clients by documenting the contracts and preserved the records for safekeeping.
- Safeguarded confidential information and assets for the clients and corporation.
- Collaborated with colleagues, stakeholders, and clients.
- Operated machines and technical equipment in a consistent and responsible manner
- Directed contracts, loaded trucks, and controlled job functions.
- Construction piece work career, around California.
- Bended sheet metal to wrap posts and window frames.

Valley Relocation, Moving Associate

January 2005 - December 2005

- Provided excellent customer service, while assisting clients with residential and industrial moving services.
- Safeguarding confidential information and assets for the clients, while collaborating with colleagues, stakeholders, and the public.
- Used forklift to move and organize crates in the warehouse.
- Efficient use of dollies and hand trucks is necessary to move objects safely.
- Commercial and Residential relocation provided around the Bay Area.